MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics | **PERIODIC TEST – 01 (POLYNOMIALS)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 2: Polynomials** | **MM: 30**

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, zero skipped steps, and final boxed answers with proper units.
2. In alternative (**OR**) sub-questions, complete step-by-step solutions and equal step marking apply to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Roots rejected by physical domain/restrictions are explicitly stated.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Objective & MCQs

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Q1. If α, β are zeroes of $f(x) = ax^2 + bx + c$, then $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$ is equal to: [1]

(A) $\frac{b^2 - 2ac}{c^2}$ (B) $\frac{b^2 - 2ac}{a^2}$ (C) $\frac{b^2 + 2ac}{c^2}$ (D) $\frac{b^2 - 2ac}{b^2}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

We know Vieta's formulas and the sum-of-squares identity ($c \neq 0$):

Formula Used: $\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}, \quad \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

Simplifying over common denominator $\alpha^2\beta^2$:

$$\frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{\alpha^2 + \beta^2}{(\alpha\beta)^2} = \frac{\frac{b^2}{a^2} - \frac{2c}{a}}{\frac{c^2}{a^2}} = \frac{\frac{b^2 - 2ac}{a^2}}{\frac{c^2}{a^2}} = \boxed{\frac{b^2 - 2ac}{c^2}}. \quad [1M]$$

Q2. What should be added to the polynomial $x^2 - 5x + 4$ so that 3 is a zero of the resulting polynomial? [1]

(A) 1 (B) 2 (C) 4 (D) 5

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B)

Let $p(x) = x^2 - 5x + 4$ and let k be added so that $q(x) = x^2 - 5x + 4 + k$ satisfies $q(3) = 0$. We know:

Formula Used: $x = 3$ is a zero of $q(x) \implies q(3) = 0$

Evaluating $p(3) = 3^2 - 5(3) + 4 = 9 - 15 + 4 = -2$. Thus $-2 + k = 0 \implies \boxed{k = 2}$. [1M]

Q3. If 2 and $-\dfrac{1}{2}$ are zeroes of the polynomial $px^2 + 5x + r$, then: [1]

(A) $p = r = 2$ (B) $p = r = -2$ (C) $p = 2,\ r = -2$ (D) $p = -2,\ r = 2$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B)* [with printed typo check]

We know Vieta's sum and product of zeroes relations for $px^2 + 5x + r$:

Formula Used: $\alpha + \beta = -\dfrac{5}{p}, \qquad \alpha\beta = \dfrac{r}{p}$

Exact calculation for printed data ($\alpha = 2$, $\beta = -\frac{1}{2}$):

$$2 + \left(-\frac{1}{2}\right) = \frac{3}{2} = -\frac{5}{p} \implies p = -\frac{10}{3}, \qquad 2\left(-\frac{1}{2}\right) = -1 = \frac{r}{p} \implies r = \frac{10}{3},$$

which misses options (A)–(D) due to a sign misprint in the second zero ($-\frac{1}{2}$ vs. $+\frac{1}{2}$). [0.5M]

Intended question (2 and $+\frac{1}{2}$, i.e., factors $(x-2)$ and $(x-\frac{1}{2})$): $2 + \frac{1}{2} = \frac{5}{2} = -\frac{5}{p} \implies p = -2$, and $2\left(\frac{1}{2}\right) = 1 = \frac{r}{p} \implies r = p = -2$, giving intended key $\boxed{\text{(B) } p = r = -2}$. [0.5M]

Q4. If α, β are the zeroes of the polynomial $x^2 - 1$, then the value of $(\alpha + \beta)$ is: [1]

(A) 2 (B) -2 (C) 0 (D) 1

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C)

Comparing $x^2 + 0 \cdot x - 1$ with $ax^2 + bx + c$, we have $a = 1$, $b = 0$. We know:

Formula Used: $\alpha + \beta = -\dfrac{b}{a}$

Hence $\alpha + \beta = -\frac{0}{1} = \boxed{0}$ (equivalently $x = \pm 1 \implies 1 + (-1) = 0$). [1M]

Q5. Assertion-Reason: [1]

Assertion (A): The polynomial $p(x) = x^2 + 4x + 5$ has no real zeroes.

Reason (R): A quadratic polynomial $ax^2 + bx + c$ has no real zeroes if $b^2 - 4ac < 0$.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) Both A and R are true and R is the correct explanation of A.

We know the discriminant criterion for real zeroes of $ax^2 + bx + c$:

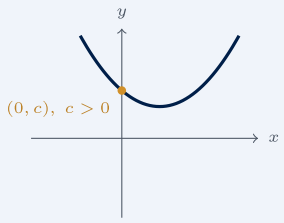
Formula Used: $D = b^2 - 4ac < 0 \implies$ No real zeroes

For $p(x) = x^2 + 4x + 5$: $D = 4^2 - 4(1)(5) = 16 - 20 = -4 < 0$ (also $p(x) = (x+2)^2 + 1 \geq 1 > 0$).

Thus both **(A)** and **(R)** are true and **(R)** explains **(A)**. [1M]

Q6. If the graph of a quadratic polynomial $ax^2 + bx + c$ cuts the positive direction of the y-axis, what is the sign of c? Justify your answer. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]



Geometric Meaning: At $x = 0$, both ax^2 and bx vanish, leaving $y = c$. Crossing above the x-axis on the y-axis forces $c > 0$.

Let $p(x) = ax^2 + bx + c$. We know that any point lying on the y-axis has abscissa $x = 0$:

Formula Used: y-intercept of $y = p(x)$ is $\big(0,\ p(0)\big) = (0,\ c)$

Substituting $x = 0$: $y = a(0)^2 + b(0) + c = c$. [1M]

Because the graph intersects the *positive* direction of the y-axis, the y-coordinate at $x = 0$ is strictly greater than 0, so $\boxed{c > 0 \text{ (positive)}}$. [1M]

Q7. Show that the quadratic polynomial $f(x) = x^2 + 2x + 5$ has no real zeroes. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Comparing $f(x) = x^2 + 2x + 5$ with $ax^2 + bx + c$, we have $a = 1$, $b = 2$, $c = 5$. We know:

Formula Used: $D = b^2 - 4ac, \qquad (x + 1)^2 \geq 0\ \forall\ x \in \mathbb{R}$

Method 1 (Discriminant): $D = 2^2 - 4(1)(5) = 4 - 20 = -16 < 0 \implies$ no real zeroes. [1M]

Method 2 (Completing the square):

$$f(x) = (x^2 + 2x + 1) + 4 = (x + 1)^2 + 4 \geq 0 + 4 = 4 > 0 \quad \forall\ x \in \mathbb{R}.$$

Hence $\boxed{f(x) \neq 0 \text{ for any } x \in \mathbb{R}}$, proving $f(x)$ has no real zeroes. [1M]

Q8. If α and β are the zeroes of the polynomial $f(x) = x^2 + x - 2$, find the value of $\dfrac{1}{\alpha} - \dfrac{1}{\beta}$. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

From $f(x) = x^2 + x - 2$, we have $a = 1$, $b = 1$, $c = -2$. We know Vieta's formulas and $(\beta - \alpha)^2$:

Formula Used: $\alpha + \beta = -1, \quad \alpha\beta = -2, \quad (\beta - \alpha)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

Factoring $f(x) = (x + 2)(x - 1)$, the zeroes are 1 and -2. Computing $|\beta - \alpha|$:

$$(\beta - \alpha)^2 = (-1)^2 - 4(-2) = 1 + 8 = 9 \implies \beta - \alpha = \pm 3. \qquad \textbf{[1.5M]}$$

Now substituting into $\frac{1}{\alpha} - \frac{1}{\beta} = \frac{\beta - \alpha}{\alpha\beta}$ (considering both orderings of $\alpha, \beta \in \{1, -2\}$):

$$\frac{1}{\alpha} - \frac{1}{\beta} = \frac{\pm 3}{-2} = \boxed{\pm \frac{3}{2}} \quad \left(\text{i.e., } \frac{3}{2} \text{ if } \alpha = 1, \beta = -2;\ -\frac{3}{2} \text{ if } \alpha = -2, \beta = 1\right). \qquad \textbf{[1.5M]}$$

Q9. If α, β are the zeroes of $f(x) = x^2 - 2x + 3$, find a quadratic polynomial whose zeroes are $\alpha + 2$ and $\beta + 2$. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

For $f(x) = x^2 - 2x + 3$, we have $\alpha + \beta = 2$ and $\alpha\beta = 3$ (note $D = -8 < 0$, so $\alpha, \beta \notin \mathbb{R}$, yet symmetric Vieta relations hold algebraically). We know:

Formula Used: $p(x) = k\left[x^2 - S'x + P'\right]$

Computing S' and P' for the new zeroes $\alpha + 2$ and $\beta + 2$:

$$S' = (\alpha + 2) + (\beta + 2) = (\alpha + \beta) + 4 = 2 + 4 = 6, \qquad \textbf{[1M]}$$
$$P' = (\alpha + 2)(\beta + 2) = \alpha\beta + 2(\alpha + \beta) + 4 = 3 + 2(2) + 4 = 11. \qquad \textbf{[1M]}$$

Taking $k = 1$, the required quadratic polynomial is $\boxed{p(x) = x^2 - 6x + 11}$. **[1M]**

Q10. If α and β are the zeroes of the quadratic polynomial $f(x) = ax^2 + bx + c$, then evaluate: **(i)** $\alpha^4 + \beta^4$ **(ii)** $\dfrac{1}{a\alpha + b} + \dfrac{1}{a\beta + b}$. [5]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

For $f(x) = ax^2 + bx + c$ ($a \neq 0, c \neq 0$), we know Vieta's formulas and the square identity:

Formula Used: $\alpha + \beta = -\dfrac{b}{a}, \quad \alpha\beta = \dfrac{c}{a}, \qquad x^2 + y^2 = (x+y)^2 - 2xy$

(i) Evaluating $\alpha^4 + \beta^4$: First find $\alpha^2 + \beta^2$:

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = \left(-\frac{b}{a}\right)^2 - 2\left(\frac{c}{a}\right) = \frac{b^2 - 2ac}{a^2}. \qquad [1M]$$

Squaring again via $\alpha^4 + \beta^4 = (\alpha^2 + \beta^2)^2 - 2\alpha^2\beta^2$:

$$\alpha^4 + \beta^4 = \left(\frac{b^2 - 2ac}{a^2}\right)^2 - 2\left(\frac{c}{a}\right)^2 = \frac{(b^2 - 2ac)^2 - 2a^2c^2}{a^4}$$

$$= \frac{b^4 - 4ab^2c + 4a^2c^2 - 2a^2c^2}{a^4} = \boxed{\frac{b^4 - 4ab^2c + 2a^2c^2}{a^4}}. \qquad [1.5M]$$

(ii) Evaluating $\dfrac{1}{a\alpha + b} + \dfrac{1}{a\beta + b}$: Note $\alpha + \beta = -\frac{b}{a} \implies a\alpha + b = -a\beta$ and $a\beta + b = -a\alpha$ (also verify via direct common denominator):

$$\frac{1}{a\alpha + b} + \frac{1}{a\beta + b} = \frac{(a\beta + b) + (a\alpha + b)}{(a\alpha + b)(a\beta + b)} = \frac{a(\alpha + \beta) + 2b}{a^2\alpha\beta + ab(\alpha + \beta) + b^2} \qquad [1M]$$

$$= \frac{a(-b/a) + 2b}{a^2(c/a) + ab(-b/a) + b^2} = \frac{-b + 2b}{ac - b^2 + b^2} = \boxed{\frac{b}{ac}}. \qquad [1.5M]$$

Q11. Find the zeroes of the quadratic polynomial $g(x) = a(x^2 + 1) - x(a^2 + 1)$, where $a \neq 0$. Verify the relationship between the zeroes and the coefficients. [5]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

Expanding $g(x)$ into standard form $Ax^2 + Bx + C$:

$$g(x) = ax^2 - (a^2 + 1)x + a \implies A = a, \quad B = -(a^2 + 1), \quad C = a. \qquad [1M]$$

We know Vieta's coefficient relations:

Formula Used: $\alpha + \beta = -\dfrac{B}{A}, \qquad \alpha\beta = \dfrac{C}{A}$

Step 1 – Factorisation: Splitting $-(a^2 + 1)x = -a^2x - x$:

$$g(x) = ax^2 - a^2x - x + a$$
$$= ax(x - a) - 1(x - a) = (ax - 1)(x - a). \qquad [1.5M]$$

Setting $g(x) = 0$ gives zeroes $\alpha = \boxed{\dfrac{1}{a}}$ and $\beta = \boxed{a}$. [1M]

Step 2 – Verification:

$$\alpha + \beta = \frac{1}{a} + a = \frac{a^2 + 1}{a}, \qquad -\frac{B}{A} = -\frac{-(a^2 + 1)}{a} = \frac{a^2 + 1}{a}, \qquad [0.75M]$$

$$\alpha\beta = \left(\frac{1}{a}\right)(a) = 1, \qquad \frac{C}{A} = \frac{a}{a} = 1. \qquad [0.75M]$$

Q12. Case Study: Roller Coaster Track Geometry

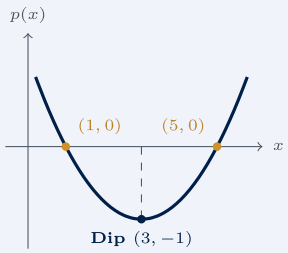
A junior roller coaster drop-and-climb track is modelled by $p(x) = \frac{1}{4}(x^2 - 6x + 5)$, where x is horizontal distance (in tens of m) and $p(x)$ is elevation relative to platform level (in tens of m). [5]

(i) Find the zeroes of the polynomial $p(x)$ and state their physical significance. **[1]**

(ii) Find the coordinates of the lowest dip point of the track and state the maximum drop depth. **[2]**

(iii) **(a)** If a safety sensor is fixed at $x = 3$, determine the elevation of the track at the sensor. **OR (b)** If a modified track profile has zeroes at $x = 2$ and $x = 8$, write its quadratic polynomial equation. **[2]**

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 5 Marks]



Track Interpretation: Since 1 unit = 10 m, zeroes $(1, 0)$ and $(5, 0)$ locate platform crossings at 10 m and 50 m, while vertex $(3, -1)$ represents a 10 m drop at $x = 30$ m.

Formula Used: $p(x) = \frac{1}{4}(x^2 - 6x + 5) = \frac{1}{4}(x - 3)^2 - 1$

(i) Zeroes & significance: Setting $p(x) = 0$:

$$\frac{1}{4}(x^2 - 6x + 5) = 0 \implies (x - 1)(x - 5) = 0 \implies \boxed{x = 1,\ 5}.$$

Physically, at horizontal distances 10 m ($x = 1$) and 50 m ($x = 5$), the track crosses exact platform elevation ($p = 0$). [1M]

(ii) Lowest dip point & max depth: Completing the square:

$$p(x) = \frac{1}{4}[(x - 3)^2 - 9 + 5] = \frac{1}{4}[(x - 3)^2 - 4] = \frac{1}{4}(x - 3)^2 - 1. \qquad [1M]$$

Since $\frac{1}{4}(x-3)^2 \geq 0$, minimum occurs at $x = 3$ with $p(3) = -1$. Coordinates of lowest dip are $\boxed{(3, -1)}$, giving a maximum drop depth of $\boxed{1 \text{ unit} = 10 \text{ m}}$ below the platform. [1M]

(iii)(a) Elevation at $x = 3$: Substituting $x = 3$:

$$p(3) = \frac{1}{4}(3^2 - 6(3) + 5) = \frac{1}{4}(9 - 18 + 5) = \frac{-4}{4} = \boxed{-1 \text{ (i.e., 10 m below platform)}}. \qquad [2M]$$

OR (Alternative Part (b) – Modified track with zeroes 2 and 8): Using $S = 2 + 8 = 10$ and $P = 2 \times 8 = 16$:

$$q(x) = k(x^2 - 10x + 16), \quad k > 0. \qquad [2M]$$

Thus the monic form is $\boxed{x^2 - 10x + 16}$, or for $k = \frac{1}{4}$, $\boxed{\frac{1}{4}(x^2 - 10x + 16)}$. **(Full credit)**

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